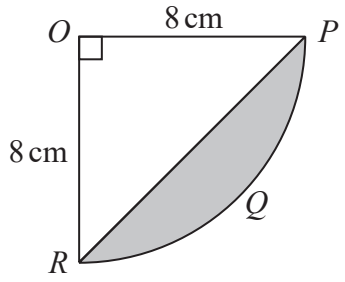
7 The diagram shows a sector $OPQR$ of a circle, centre O and radius 8 cm.



OPR is a triangle.

Work out the area of the shaded segment PQR.
Give your answer correct to 3 significant figures.

Right angle at O gives a quarter-circle sector.

Sector = $(1/4)\pi(8^2) = 16\pi = 50.265...$

Triangle OPR = $(1/2)(8)(8) = 32$

Segment = $16\pi - 32 = 18.265...$

= 18.3 cm^2 (3 s.f.)

1 mark
triangle area 32

1 mark
circle area $\pi \times 8^2$

1 mark
quarter-sector area 16π

1 mark
18.3 cm^2

..... 18.3 ..... cm^2

(Total for Question 7 is 4 marks)

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